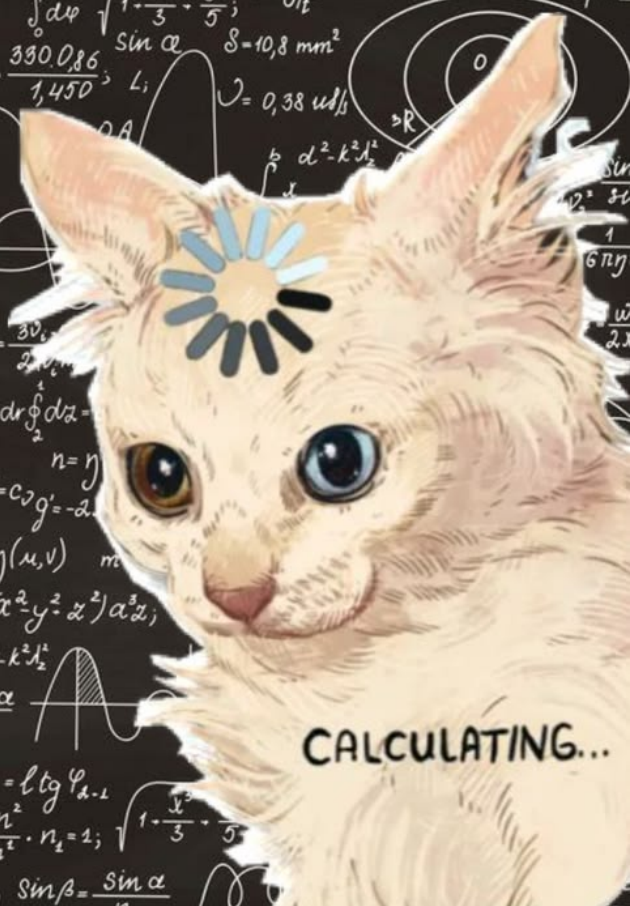


COMPOSITION BOOK

Machine Learning

100 SHEETS

9.75 X 7.5 IN / 24.76 X 19.0 CM



CALCULATING...

Introductions

Definition:

A computer program is said to learn from **EXPERIENCE E** on a class of **TASK T** and **PERFORMANCE MEASURE P**, if its skill on task T (measured by P) improves with experience E.

A program could learn to play checkers and improve its performance (win rate) by playing games against itself.

In general, to have a well-defined learning problem, we must identify three elements:

1. **CLASS OF TASKS (T)**
2. **PERFORMANCE MEASURE TO IMPROVE (P)**
3. **THE SOURCE OF EXPERIENCE (E)**

Example checkers

Let's take the game of **CHECKERS** as an example where:

1. the **TASK T** is playing checkers
2. the **PERFORMANCE MEASURE P** is the percentage of games won in tournaments
3. the **EXPERIENCE E** comes from games played against itself.

To make the program learn, we need to decide what knowledge it should acquire, how to represent it, and which learning algorithm to use.

The program needs to choose the best move

how?

Instead of directly predicting every move with a **COMPLEX FUNCTION (CHOOSE MOVE)** ✗

it is much easier to use an evaluation function **V** ✓

An **EVALUATION FUNCTION V** that assigns a numerical value to each board position.

The higher score, the more advantageous that position is.

Ideally, this function gives +100 to a winning board, -100 to a losing board and 0 to a draw.

